Robotics Programme: Worksheet 1

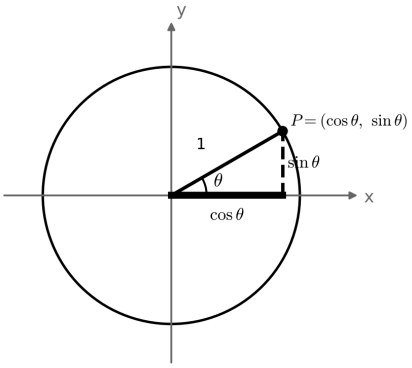
Alexander | Session 1: Meet the D1 Edu

Note: use g = 9.81 m/s² and give numbers to three significant figures unless a question says otherwise. The tables below hold every formula and value needed.

Formula	Meaning
$W = mg$	Weight in newtons (N), mass m in kg
$E_k = \frac{1}{2}mv^2$	Kinetic energy in joules (J), speed v in m/s
$\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180}$	Degrees to radians (multiply by 180/π to go back)
$\cos^2\theta + \sin^2\theta = 1$	On the unit circle cos is the x-coordinate and sin is the y-coordinate

θ	0°	30°	45°	60°	90°	180°
cos θ	1	0.866	0.707	0.5	0	-1
sin θ	0	0.5	0.707	0.866	1	0

Value	Number
Mass of robot, with accessory	17.2 kg
Joint ranges	abad -28° to 28°, hip -170° to 66°, knee 35° to 156°
Body frame	X forward, Y left, Z up



Radius 1. Here $\theta = 30°$, so cos = 0.866 (thick line, sideways) and sin = 0.5 (dashed line, upwards).

Unit circle at 30 degrees.

1. Find the weight of the robot with the accessory (17.2 kg) in newtons. 17.2*9.81=169 newtons

2. Find the kinetic energy of the robot with accessory at 2.0 m/s. 1/2*17.2*2^2=4/2*17.2=34.4 Joules

3. A robot with 8 legs has 3 joints per leg. (a) How many actuated joints? (b) How many numbers describe its full configuration, counting the body's position and orientation? a: 3*8=24 b: 24+6=30

4. Convert 156° to radians, -170° to radians, and 0.6109 rad to degrees (multiply by 180/π).

5. Without a calculator, use the reference table: (a) cos 60° and sin 60°. (b) At 180°, where on the circle is the point, and what are cos and sin? (c) A point on the unit circle has cos θ = 0.6. Find sin θ, taking it positive.

a. cos60=1/2 sin60=sqrt3/2 b. cos180=-1 sin180=0 (-1,0) c. 0.8

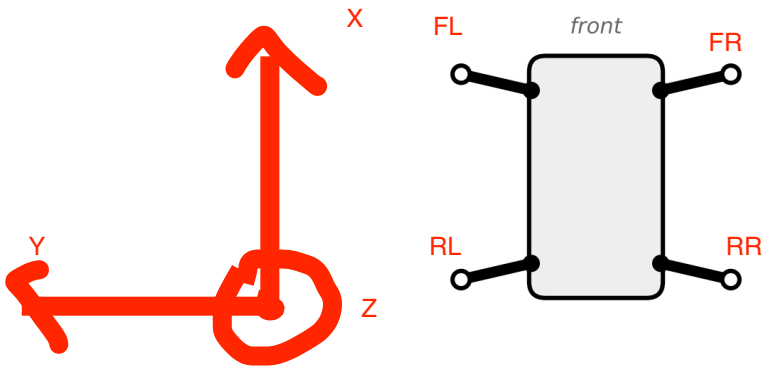
6. Open GeoGebra Classic at https://www.geogebra.org/classic. If a menu covers the graph, click the graph. Find the Input row (top of the Algebra panel in a wide window, below the list in a narrow one). Type these lines, pressing Return after each:

```
a=Slider(0,2pi,0.01)
P=(cos(a),sin(a))
c=Circle((0,0),1)
cx=cos(a)
sy=sin(a)
```

Then type a=pi/3 in the Input row. Write down cx and sy and say whether they match question 5(a). Then type a=2 and write down P.

a=pi/3, cx=0.5, sy=0.87
a=2, P=(-0.42,0.91)

7. The figure is a top view of the D1. Print or copy it. Mark the X, Y and Z axes (X forward, Y left, Z up) and label the legs FR, FL, RR and RL.



Top view of the D1 (not to scale).

8. The robot is turned by ψ anticlockwise from the world x axis and walks with body velocity (vx, vy). The world velocity is:

$$x_w = v_x \cos\psi - v_y \sin\psi$$

$$y_w = v_x \sin\psi + v_y \cos\psi$$

Find the world velocity when: (a) ψ = 45°, vx = 1.0 m/s, vy = 0. (b) ψ = -90°, vx = 0.6 m/s, vy = 0.8 m/s. (Use cos(-90°) = 0 and sin(-90°) = -1.) Say in words which way the robot moves.

9. Write a Python program that prints the angles 0, 30, 45, 90 and 180 degrees in radians, rounded to 4 decimal places. Run it and write down the output. Hint: import math, then math.radians(d).

Links: GeoGebra Classic and AgiBot D1 Edu SDK repository (some documents are in Chinese, so use a browser translator).

8a. x=1.0*cos45-0*sin45=(sqrt2)/2. y=1*sin(45)+0*(cos45)=(sqrt2)/2.
The robot's trajectory is the angle bisector between the x and y axis, its speed is 1 m/s.

8b. x=0.6*cos(-90)-0.8*sin(-90)=0.6*0+0.8=0.8. y=0.6*sin(-90)+0.8*cos(-90)=-0.6+0=-0.6
The robot's trajectory, when drawn from a top-down view, is 37 degrees below the x-axis. Speed is 1 m/s

9.

```
import math
angles_degrees=[0,30,45,90,180]
for d in angles_degrees:
    radians=round(math.radians(d),4)
    print(radians)
```

OUTPUT:
0
0.5236
0.7854
1.5708
3.1416